Yibo Yuan

Research Assistant at Westlake University | Biomedical Engineer | Social Neuroscience x Artificial Companions
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RESEARCH PROFILE

Biomedical engineer working across systems and computational neuroscience, embodied AI, HCI, and neuroengineering. I study whether artificial social partners can regulate social need - from hypothalamic population dynamics and naturalistic behavior to biomimetic robotic partners and authority-aware long-term AI memory.

EDUCATION

Xi'an Jiaotong University, Xi'an, China — **Sep 2021 - Jun 2025**
B.Eng. in Biomedical Engineering, School of Life Science and Technology

- Selected coursework: Pattern Recognition and Machine Learning (85/100); Biomedical Sensing and Instrumentation (84/100); Data Structures and Algorithms (80/100); Medical Device R&D Management and Product Certification (100/100).

RESEARCH APPOINTMENTS

Westlake University, Hangzhou, China — **Aug 2025 - Present**
Full-time Research Assistant, Professor Ding Liu's lab

Illinois Institute of Technology, Remote — **Nov 2025 - Present**
Remote Research Collaborator, Trustworthy and Intelligent Machine Learning Lab, Prof. Ren Wang

Westlake University, Hangzhou, China — **Mar 2026 - Present**
Cross-lab Research Collaborator, AI for Scientific Simulation and Discovery Lab, Prof. Tailin Wu

RESEARCH EXPERIENCE

Westlake University - Professor Ding Liu's lab, Hangzhou, China — **Aug 2025 - Present**
Research Assistant — ***Social Need and Hypothalamic Population Dynamics***

- Develop an end-to-end workflow linking long-duration social video, frame-level interaction labels, miniscope calcium imaging, and session-aligned neural population activity.
- Independently built an open-source behavioral-video pipeline for irregular arenas, multi-region tracking, perspective correction, temporal alignment, quantitative exports, and segmentation-based contact detection.
- Apply hazard-based behavioral analysis, latent-state modeling, neural-population geometry, Ridge readout, CKA, subspace alignment, gain modulation, and time-shift permutation controls to test how isolation and reunion reshape social-state representations; quantitative claims remain under internal verification.
- Contributed to miniscope calcium-imaging workflows and the published JoVE protocol listed below; trained in stereotactic injection, optogenetic manipulation, perfusion, and brain cryosectioning.

Westlake University x Beijing Institute of Technology, Hangzhou / Beijing — **Jan 2026 - Present**
Project originator and cross-institutional coordinator — ***Programmable Biomimetic Robotic Mouse***

- Conceived the research program and initiated the collaboration to test whether an artificial embodied partner can regulate social need and where equivalence with a biological partner fails.
- Designed a closed-loop architecture combining overhead pose perception, programmable motion, and tunable tactile, thermal, and olfactory cues; collaborated with the robotics team on hardware iteration and interface requirements.
- Completed a descriptive fixed-robot pilot analysis using semantic segmentation, calibrated distance-to-mask measures, approach bouts, immobility proxies, and rapid-retreat proxies; findings are exploratory because of limited sessions and confounded recording order.

Westlake University - Professor Ding Liu's lab, Hangzhou, China — **Aug 2025 - Present**
Research tool developer — ***Experimental Platforms for Behavioral Neuroscience***

- Built and open-sourced an eight-channel STM32/TTP224 capacitive lickometer with UART transmission and a Python interface for real-time event visualization and CSV storage.
- Designed and assembled a programmable three-axis stepper-motor motion platform with configurable motion profiles; modeled and 3D-printed custom experimental components.

Illinois Institute of Technology - TIML Lab, Remote — **Nov 2025 - Present**
Remote Research Collaborator, supervised by Prof. Ren Wang — ***Cross-subject EEG-fNIRS Learning and Reliable Fusion***

- Built participant-disjoint and leave-one-subject-out evaluation pipelines for cross-subject EEG/fNIRS classification, few-shot adaptation, probability calibration, and multimodal fusion.
- Developed and audited heterogeneous temporal, Riemannian, and hemodynamic experts; on the 26-subject Shin2018-WG benchmark, the frozen three-expert fusion reached 74.0% participant-equal balanced accuracy under a K=5-per-class protocol.
- Evaluated learned task-prior correction and meta-adaptive schedules; preserved negative results showing that calibration gains do not automatically imply accuracy gains and that weak modalities can degrade fusion.

Westlake University - AI for Scientific Simulation and Discovery Lab, Hangzhou, China **Mar 2026 - Present**
Cross-lab Research Collaborator, supervised by Prof. Tailin Wu ***Physics-informed fMRI Dynamics***

- Contributed to HCP task-fMRI preprocessing, multimodal event-regressor integration, V-JEPA/MAE training refactoring, SIREN/INR temporal modeling, and vectorized multiscale alignment for large-scale brain-dynamics modeling.
- Applied vortex-pattern detection and turbulence-oriented visualization to resting-state fMRI, connecting spatial power-law analysis, Hopf whole-brain models, and latent-dynamics reconstruction; distributed FSDP2 migration and training validation are ongoing.

Westlake University - Professor Xiaodong Liu collaboration, Hangzhou, China **2026 - Present**
Behavioral-video analysis contributor ***Constrained 2.5D Analysis of Parkinsonian Macaque Behavior***

- Developed SAM2 mask-based body tracking, cage calibration, separate front- and floor-plane trajectory proxies, contact-state handling, activity fragmentation, and spatial-occupancy summaries; current n=2 findings are reported descriptively without frame-level pseudoreplication.

Westlake University - Protein Aggregation and Degradation Lab, Hangzhou, China **Feb 2026 - Mar 2026**
Independent developer, supervised by Prof. Rui Fang ***Neuron Axon Growth Tracking Tool***

- Built an OpenCV/Tkinter GUI that traces axon growth from user-specified waypoints using brightness, direction, smoothness, step-size, and boundary constraints; exports calibrated trajectories and growth-speed tables.

INDEPENDENT AND COLLABORATIVE AI RESEARCH

Mori: Authority-Calibrated Family Companion Agent, Independent project **Aug 2026 - Present**
Project lead and full-stack developer ***Live research prototype***

- Designed and deployed a multi-person family-agent prototype that separates technical capability, delegated authority, and accepted duty, routing decisions through ACT, ASK, DEFER, HANDOFF, or ABSTAIN.
- Implemented a Mandate Critic - Actor - Grounding Reviewer prompt chain, per-person four-level authority controls, and purpose-tagged memory policies for retaining, sharing, analyzing, interpreting, advising, intervening, and external connection.
- Built the React/TypeScript and Cloudflare system with server-side model integration; manually exercised boundary-sensitive scenarios. Automated benchmarks and user studies are planned, not yet completed.

TAS-Gate: Authority-aware Long-term Memory, Independent research **Aug 2026 - Present**
Researcher ***Research proposal and experimental design***

- Formulated a structured update rule in which temporal scope, authorization semantics, and speaker qualification must all hold before language can change an AI's binary action authority.
- Designed counterfactual NLP tasks, ablations, long-dialogue tests, and a two-layer memory architecture in which experiential success and trust cannot directly expand rule-level authority; implementation and evaluation are ongoing.

Connectome-constrained Meta-learning, Westlake collaboration **Mar 2026 - Present**
Research idea contributor ***Drosophila central-complex topology***

- Proposed an evolution-as-meta-learning framework and explored FlyWire/NeuPrint connectome extraction, Brian2/Nengo simulation, topology-matched controls, and navigation-oriented tests; authorship, venue, and confirmatory results remain open.

UNDERGRADUATE RESEARCH AND ENGINEERING

Xi'an Jiaotong University, Xi'an, China **Jan 2025 - Jun 2025**
Graduation project, supervised by Profs. Xiang Chen and Jin Li ***Portable Bioimpedance Spectroscopy System***

- Designed and bench-tested a portable tissue-water-content sensing prototype based on multi-frequency bioimpedance spectroscopy and the Cole-Cole model, integrating STM32 control, an AD5933 front end, a two-electrode circuit, PCB design, and hardware debugging.

Xi'an Jiaotong University, Xi'an, China **Mar 2024 - Sep 2024**
Project leader, supervised by Prof. Jin Li ***Eight-channel EEG Acquisition System***

- Led PCB architecture and circuit design for an ADS1299/STM32F103 system, including power/reference, input protection, anti-aliasing, bias drive, and test-signal paths; completed power/reference/clock checks and specified the remaining noise, CMRR, communication, and signal-validation plan.

Institute of Mitochondrial Biomedicine, Xi'an Jiaotong University, Xi'an, China **Jan 2022 - Oct 2022**
Laboratory member, supervised by Prof. Dan Tan ***Synthetic E. coli for soil improvement***

- Supported plasmid design and optimization, PCR/RT-PCR, gel extraction, nucleic-acid purification, Western blotting, electrophoresis, and literature analysis for the 2022 iGEM project.

PUBLICATION AND RESEARCH OUTPUTS

Published

Dai, Y. Y., Wei, C., Yuan, Y., Rahman, M. M., & Liu, D. (2026). In vivo calcium imaging with a miniaturized microscope in the hypothalamus for understanding social behaviors in mice. *Journal of Visualized Experiments*, (229), e70401. doi:10.3791/70401

Working research outputs - status stated conservatively

- Social-need neural population dynamics: analysis and figure package in internal development; sample mapping, quantitative claims, data-use permission, and authorship must be frozen before external submission.
- EEG-fNIRS fusion and meta-adaptation: internal author-check manuscripts and audited result ledgers exist; external venue and author order remain to be confirmed with the PI.
- Mori / TAS-Gate: live prototype plus research proposal; no completed user study, system benchmark, or accepted manuscript is claimed.

OPEN-SOURCE RESEARCH SOFTWARE

Mouse-trajectory-tracking: Long-duration multi-animal tracking, custom ROIs, temporal/perspective alignment, frame-level outputs, and experimental contact detection. GitHub: github.com/BoBo1529707515/Mouse-trajectory-tracking

Lickometer: Eight-channel STM32 firmware and Python acquisition interface for capacitive lick detection. GitHub: github.com/BoBo1529707515/lickometer

NeuronTracker: GUI-assisted axon-growth tracing with biologically motivated path constraints and calibrated quantitative exports. GitHub: github.com/BoBo1529707515/Neural-tract-tracing

Mori: Authority-calibrated family-agent prototype with multi-person mandates and purpose-limited memory. Demo: mori-family-companion.pages.dev

PROFESSIONAL EXPERIENCE

Zhen Tec Technology Inc., Xi'an, China **Jun 2024 - Jul 2024**

EEG Data Analyst

- Designed IIR/FIR filters, analyzed time- and frequency-domain EEG features with MNE, computed Higuchi fractal dimension and Hjorth parameters, and supported PCA/ICA and CNN-LSTM fatigue-model development.

Collegeland Education Co. Ltd., Remote / Xi'an **May 2024 - Sep 2024**

AI Assistant Developer

- Built web-crawling and data-processing tools for education analytics and supported prompt engineering, hallucination handling, and adversarial sensitive-term testing for educational LLM workflows.

First Affiliated Hospital of Xi'an Jiaotong University, Xi'an, China **May 2023 - Jun 2023**

Clinical Laboratory Intern

- Observed rehabilitation and medical-imaging workflows, assisted routine clinical tasks, and evaluated usability constraints in rehabilitation and portable imaging equipment.

HONORS AND AWARDS

- National College Students' Innovation and Entrepreneurship Training Program, 2024 - deubiquitinating enzymes, prostate-cancer proliferation, and immune mechanisms.
- Second Prize, Innovation Track, 35th Tengfei Cup Innovation and Entrepreneurship Competition, Xi'an Jiaotong University, 2024.
- National College Students' Innovation and Entrepreneurship Training Program, 2023 - age-related changes in the human-brain single-cell atlas.
- Gold Prize, Entrepreneurship Track, 34th Tengfei Cup, Xi'an Jiaotong University, 2023.
- Second Prize, Xi'an Jiaotong University Mathematical Modeling Competition, 2023.
- Gold Medal, International Genetically Engineered Machine Competition (iGEM), 2022.

LEADERSHIP AND SERVICE

Medical and Engineering Innovators, Xi'an Jiaotong University **Jan 2024 - 2025**

Council member / External relations

- Led alumni liaison for biomedical and electrical engineering members, supported a Technology and Finance forum, and proposed a hospital-university translational project on an intranasal sustained-release device.

The Micro Light Tutoring Association, Remote **Jan 2023 - Apr 2023**

Volunteer lecturer

- Provided free one-on-one online tutoring to high-school students in Tibet.

TECHNICAL SKILLS

Programming and ML: Python, C, MATLAB, PyTorch, scikit-learn, pandas, NumPy, MNE, OpenCV, MMSegmentation, OpenMMLab, CEBRA, React, TypeScript

Neural and behavioral analysis: Miniscope calcium imaging, neural manifolds, population decoding, representational similarity, hazard analysis, long-duration tracking, semantic segmentation, SAM2, SLEAP

Neuroengineering and hardware: STM32, ADS1299, AD5933, UART, PCB design, soldering/debugging, stepper-motor control, Blender, 3D printing, experimental synchronization

Wet laboratory: Mouse stereotactic surgery, injection, perfusion, brain cryosectioning, optogenetic preparation, DNA/RNA extraction, PCR/RT-PCR, Western blotting

Research tools: Git/GitHub, Linux, Cloudflare, Bonsai, Miniscope GUI, SnapGene, Adobe Illustrator, SPSS

Languages: Mandarin Chinese (native); English (professional working proficiency)